SPARQL UniProt.RDF

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Swiss Institute of Bioinformatics

Everyone has had some introduction slash knowledge of RDF.

Tutorial plan

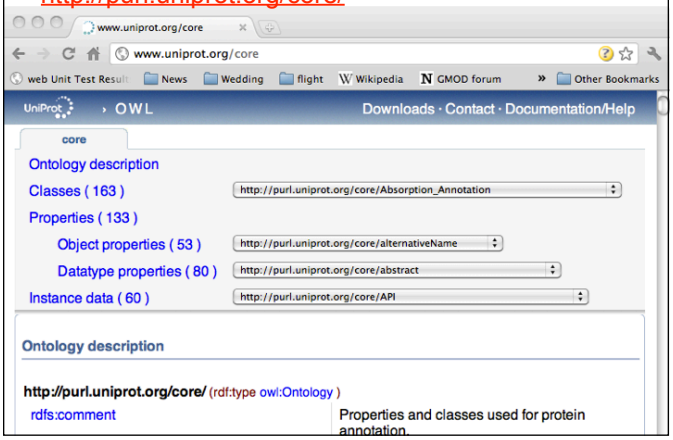
- Set up Topbraid Composer
  - Skipped in talk
- Gather data from uniprot website
- Learn sparql

You do not need Topbraid Composer to use UniProt RDF data or do sparql queries.
- Just used to tie in with morning tutorial by *J. Phil Brooks*

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You should have used Topbraid composer in this mornings tutorial.
If not have a look at next few slides.

Before starting have a look at http://purl.uniprot.org/core/



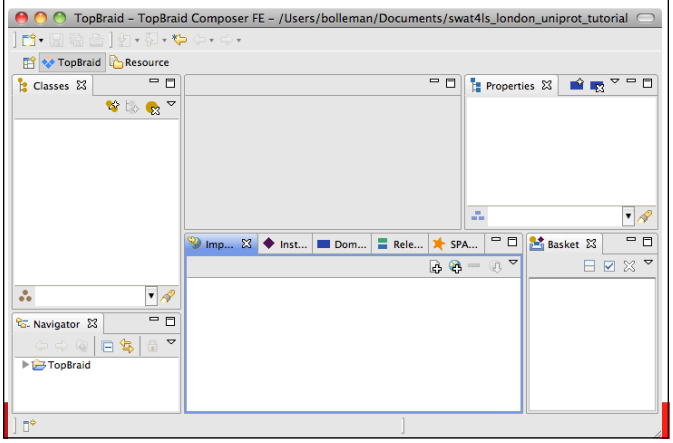
You can find the documentation on the “schema” of uniprot rdf here.

Download and install Topbraid composer

- Requirements
  - Sun/Oracle JVM
- Go to
  - http://www.topquadrant.com/products/TB_download.html
  - Register
  - Select any edition, free is ok for today

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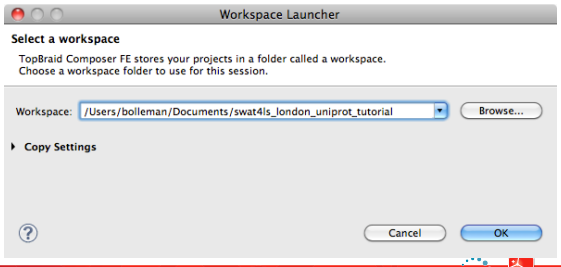
Start Topbraid



If you never used topbraid before you will have an empty workspace.

Setting up a workspace for this tutorial

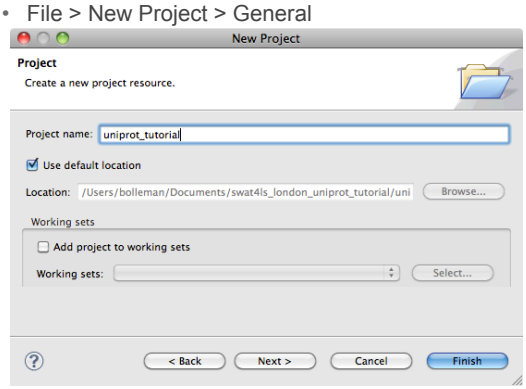
- http://www.topquadrant.com/products/TB_download.html



For today please create a new empty workspace that does not influence your previous work

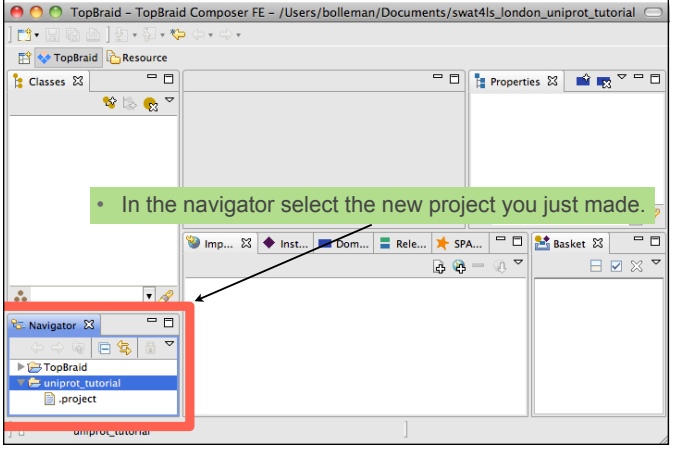
New project

- File > New Project > General



Give your project a recognizable name

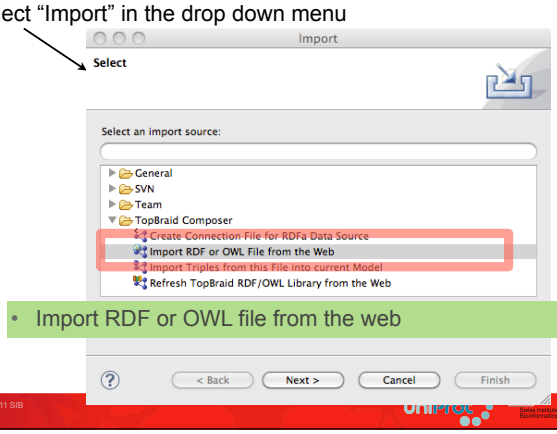
Gather data from uniprot.org website



The .project file contains the project details do not delete it!

Gather data from uniprot.org website

Right click on your new project.
Select “Import” in the drop down menu

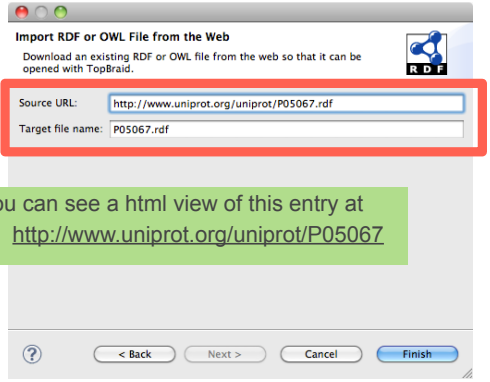


- Import RDF or OWL file from the web

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In this case we are going to use an uniprot entry for our examples.

Gather data from uniprot.org website



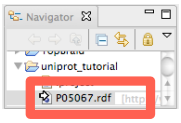
You can see a html view of this entry at
http://www.uniprot.org/uniprot/P05067

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Fill in the source and target url. Click finished
Do the same for http://www.uniprot.org/owl/core.rdf and name it core.owl
core.owl contains the "schema" data for uniprot rdf.

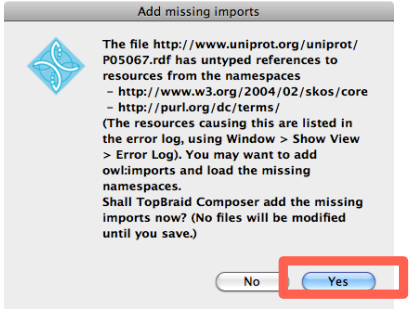
Gather data from uniprot.org website

- Open the P05067 file by double clicking



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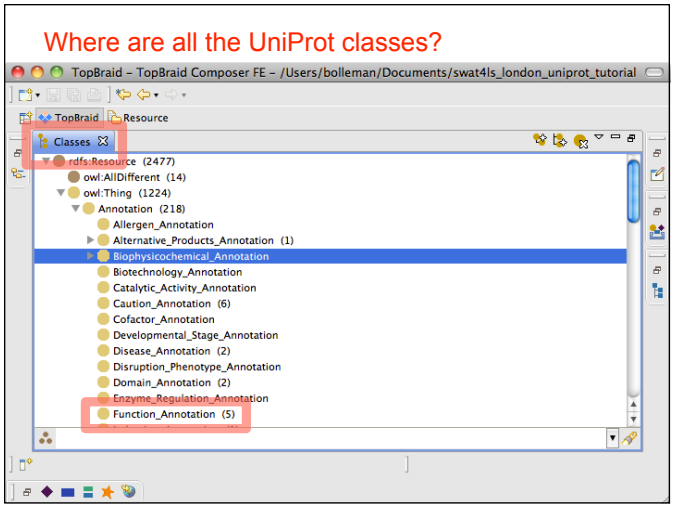
You get a very helpfull dialog.
Hit yes



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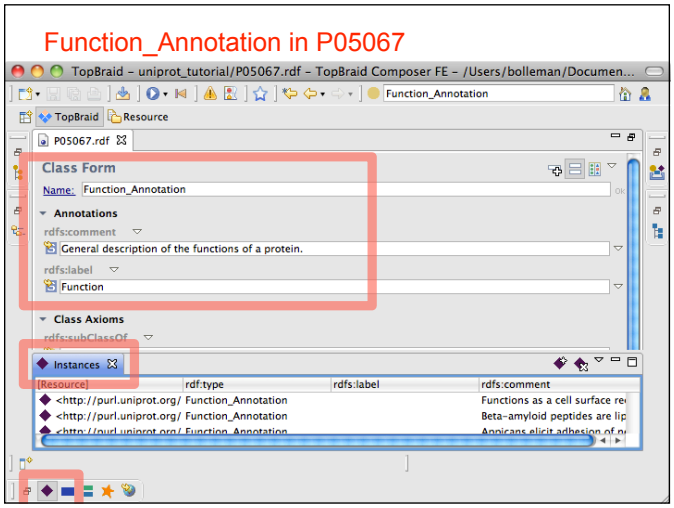
This auto imports ontologies used by uniprot that are not inside the core.owl file.
And their imports as well.

Where are all the UniProt classes?



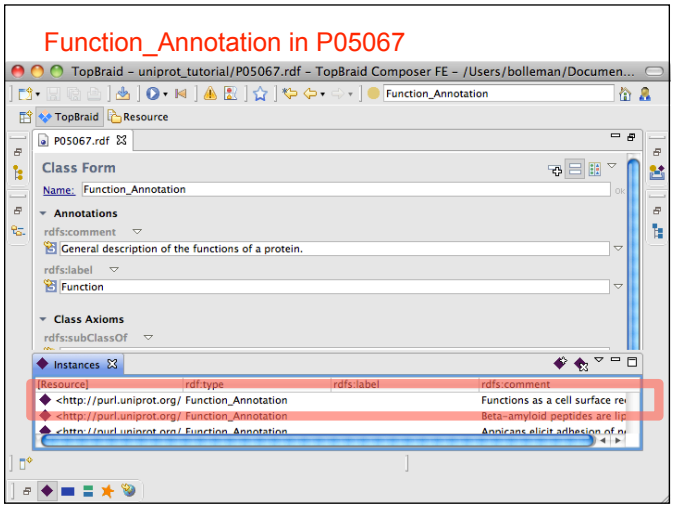
Have a look at the Tab classes. The number between the brackets is the instances of that class in your file.

Function_Annotation in P05067



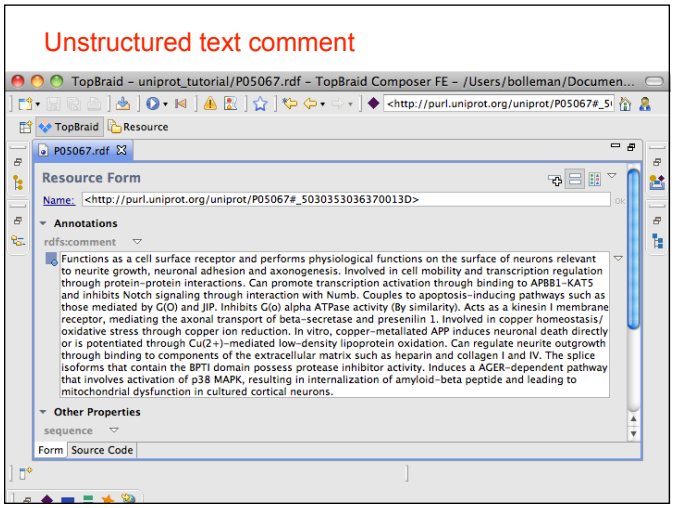
Some datatype documentation
If instance is empty double click on the Function_Annotation in the classes view.

Function_Annotation in P05067



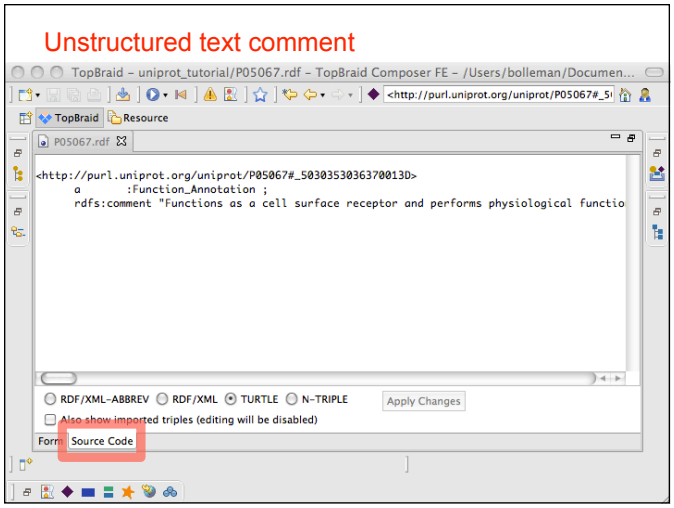
Double click on the top triple Resource to see it in more detail.

Unstructured text comment



This is the top Function_Annotation Instance of the last page.

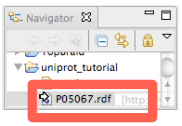
Unstructured text comment



Use the source code tab to see the triples in RAW formats.
The turtle view is helpfull when you start to write SPARQL queries.

Let's infer

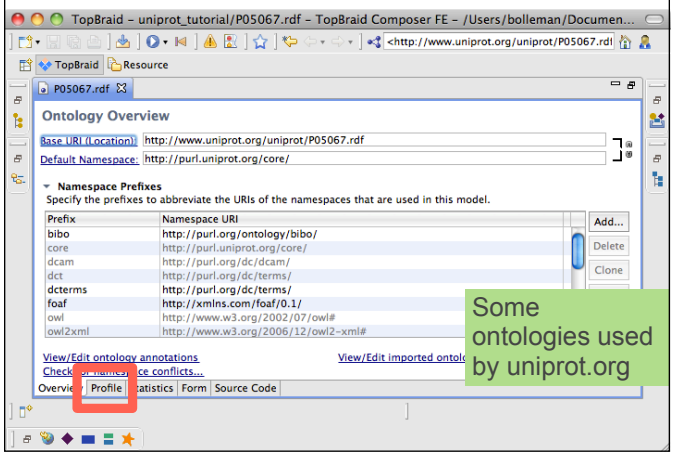
- Go back to the top level of the file by double clicking again on the file name in the navigator tab.



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You should get a view that you saw earlier in this tutorial.

Let's infer

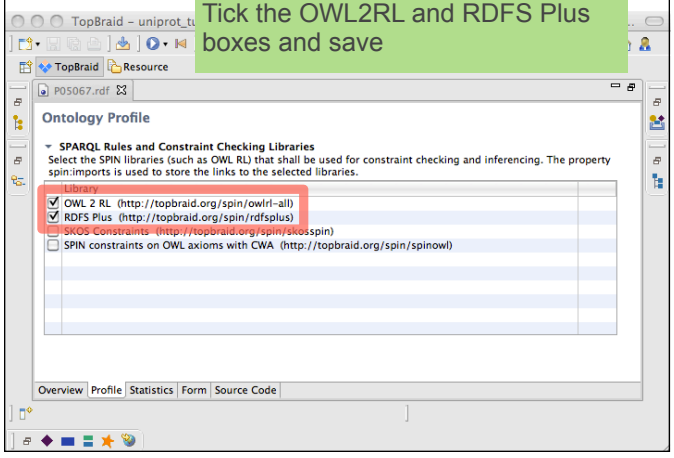


Some ontologies used by uniprot.org

Change to the profile tab

Profile tab

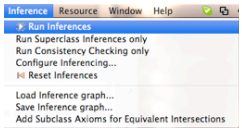
Tick the OWL2RL and RDFS Plus boxes and save

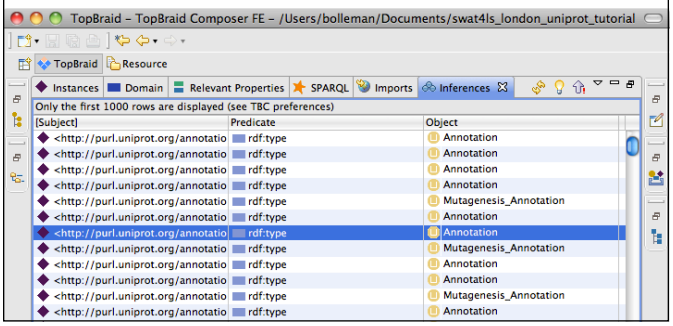


This enables the reasoner.

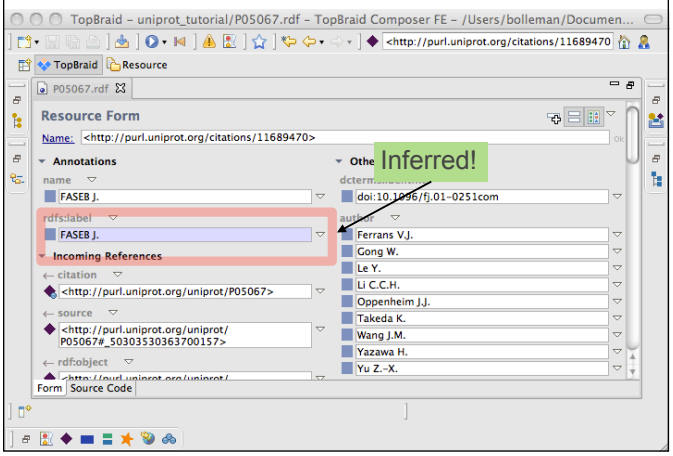
Run the reasoner

- In the menu “Inference” > select the option “Run inferences”





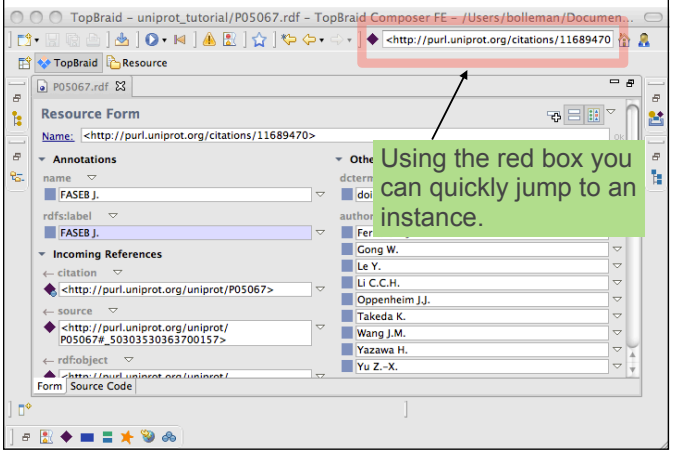
name is inferred to be a rdfs:label



Inferring can help make queries easier. Or they can trully infer new knowledge.

Side note Annotations (as above the name) are annotations in the OWL sense not in the biological curated annotation sense.

name is inferred to be a rdfs:label



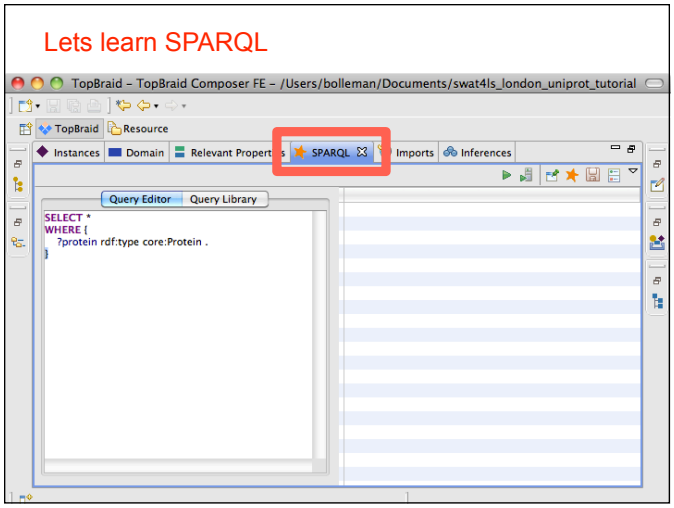
Quick navigation.

Lets learn SPARQL

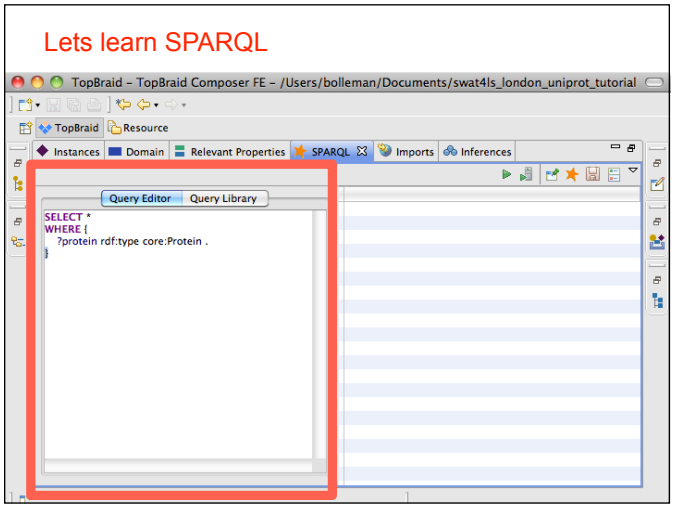
- Queries over RDF data.
  - Four basic types
    - SELECT
      - Returns “tab delimited” results
    - CONSTRUCT
      - Makes new triples
    - DESCRIBE
      - Returns all triples mentioning a resource
    - ASK
      - Return true if anything matches

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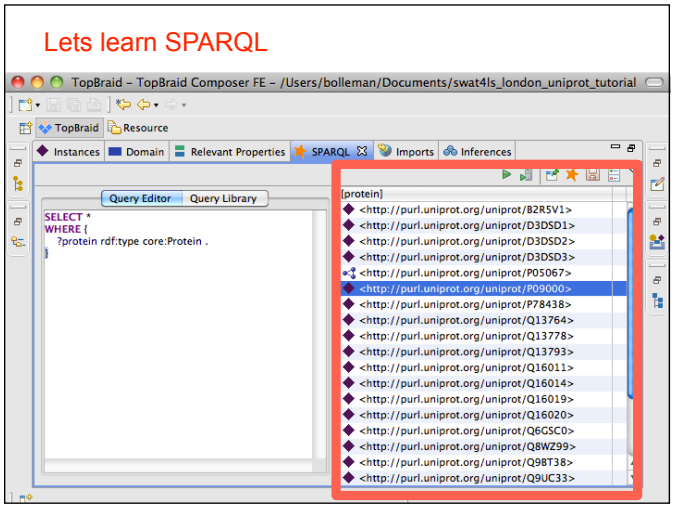
In this example session I will only show SELECT and CONSTRUCT



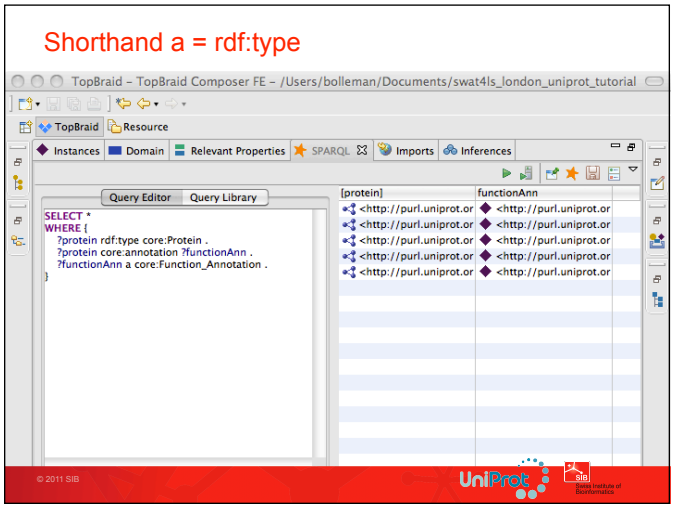
All



This is where you type your query.



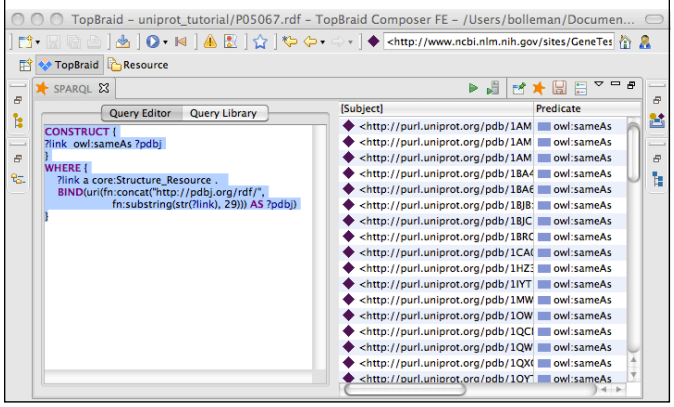
This is where you see your results.
Each line in the where clause is a triple pattern where things that start with ? are variables



Here we select those 5 instances that we saw earlier on in the classes -> instances tab

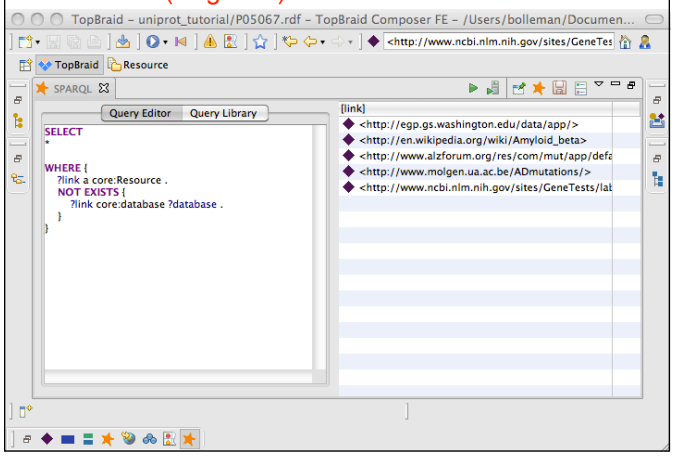
```
SELECT *
WHERE  { ?protein rdf:type core:Protein .
?protein core:annotation ?functionAnn .
?functionAnn a core:Functio_Annotation .
}
```

Constructing an owl:sameAs between two URI



str() to change a IRI into a string
concat and substring to do string manipulation
IRI() to change the string back into a IRI

Not exists (Negation)

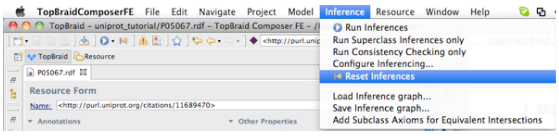



```
SELECT *
WHERE {
        ?link a core:Resource .
        NOT EXISTS {
                ?link core:database ?database .
        }
}
```

Inferencing changes the results of queries

```
SELECT *
WHERE {
   ?subject rdfs:label "FASEB J." .
}
```

Try this query before and after “reseting inferences”
In the menu bar under inference



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More uniprot rdf

- http://www.uniprot.org/downloads
  - (See bottom of page for RDF)
- http://www.uniprot.org/faq/28
- Queries on the website can be downloaded as RDF
  - e.g. only human entries
  - http://www.uniprot.org/uniprot/?query=organism%3a9606&sort=score&format=rdf

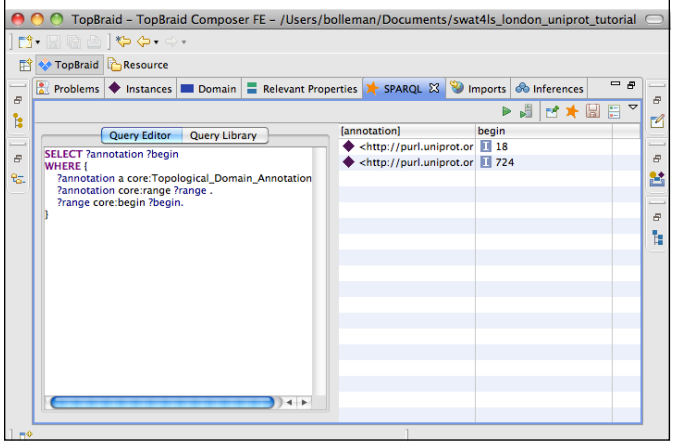
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Thank you for your time!

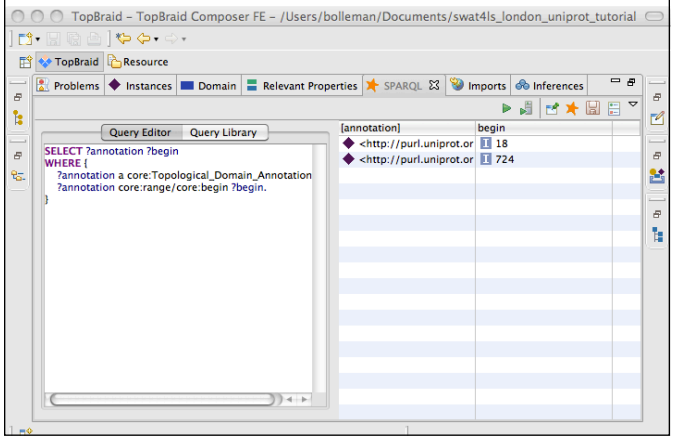
UniProt

SIB Swiss Institute of Bioinformatics

Extra material:path queries

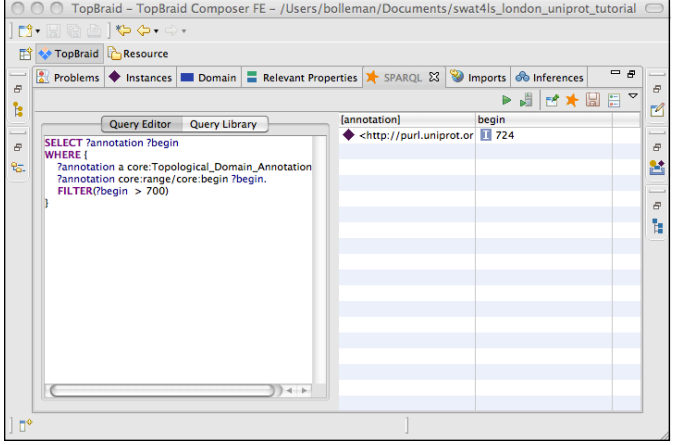


Extra material:path queries



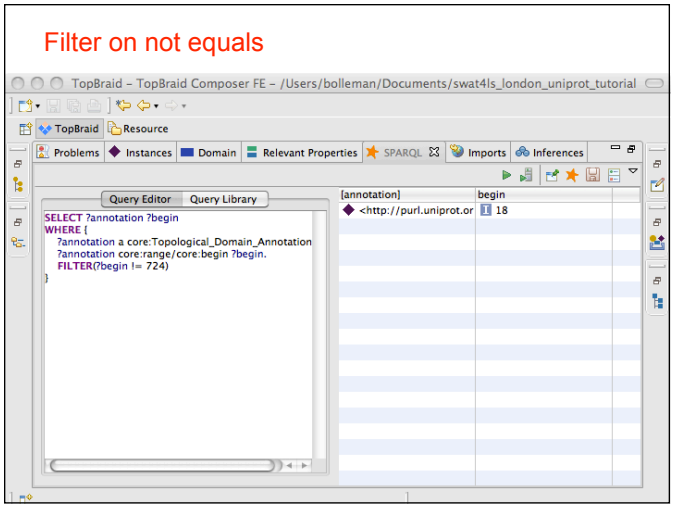
?s core:range/core:begin ?o;range property then begin property
?s core:beginlcore:end ?o; begin or end property
?s core:range* ?o;zero or more steps
?s core:range+ ?o;one or more steps
?s core:range{2,3} ?o;two or three steps
?s core:annotation/core:range/core:begin ?p any annotations begin position.

Filter



FILTER can be used to remove potential matches from the pattern.

Filter on not equals



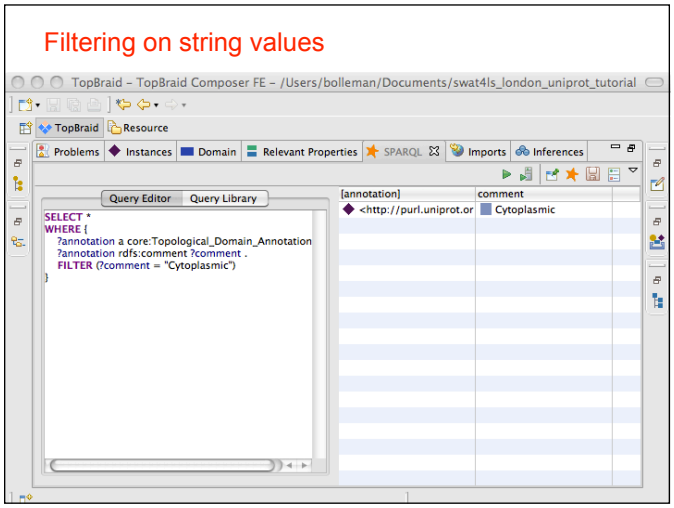
?a > ?b : a greater than b
?a < ?b : a smaller than b
?a = ?b : a same value as b
?a != ?b : a different value than b

Filters

- Options depend on the values
  - e.b. < > only work on numbers

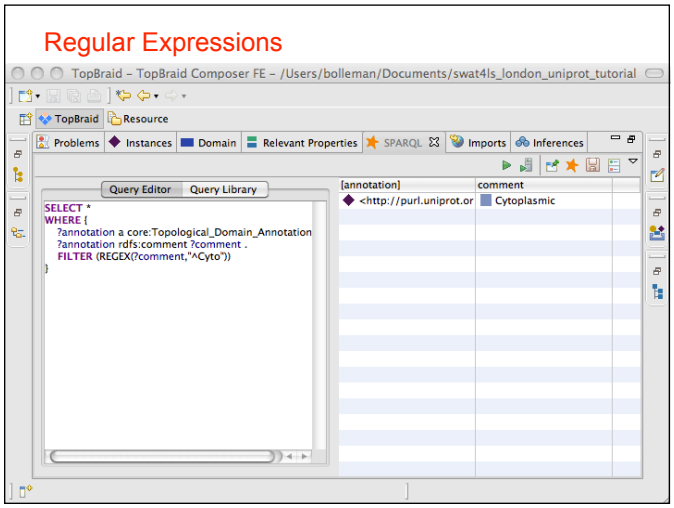
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Filtering on string values



?a = ?b : a same value as b
?a != ?b : a different value than b

Regular Expressions



Most “perl style regex options” work except for capturing groups

Why don't these queries work on the web?

- PREFIX
  - Topbraid composer uses the prefixes defined in the files "overview" tab.
  - On the web you often have to add these.

```
PREFIX :<http://purl.uniprot.org/core/>
SELECT ?x
FROM <http://purl.uniprot.org/taxonomy/>
WHERE {?x a :Taxon}
```

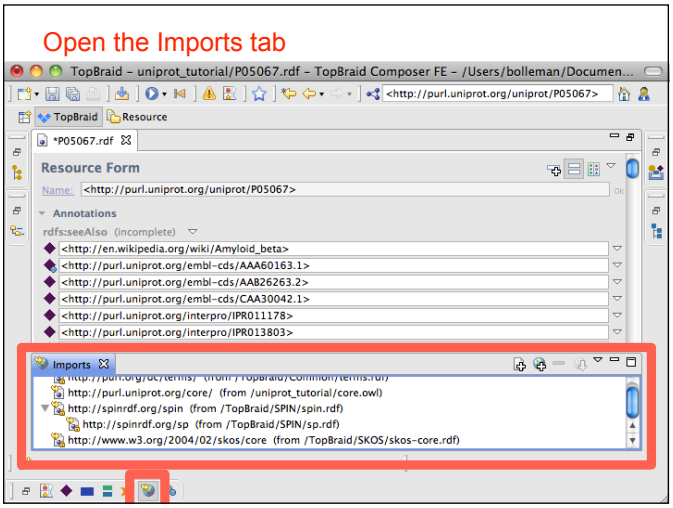
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Adding your own rules to the inferencer

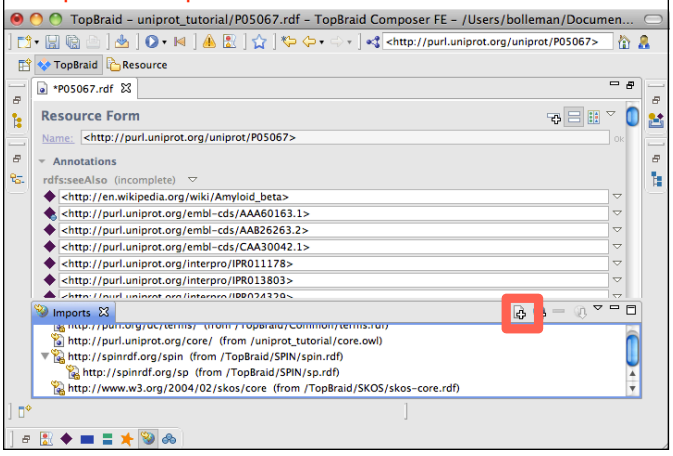
- Remember the linking between UniProt and PDBj identifiers?
- Using SPIN rules one can do this "automatically"
- First import the SPIN "schema"

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Open the Imports tab

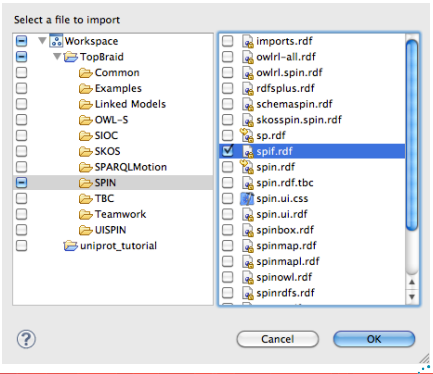


Open the Imports tab



Use the local import function to import the SPIN "schema"

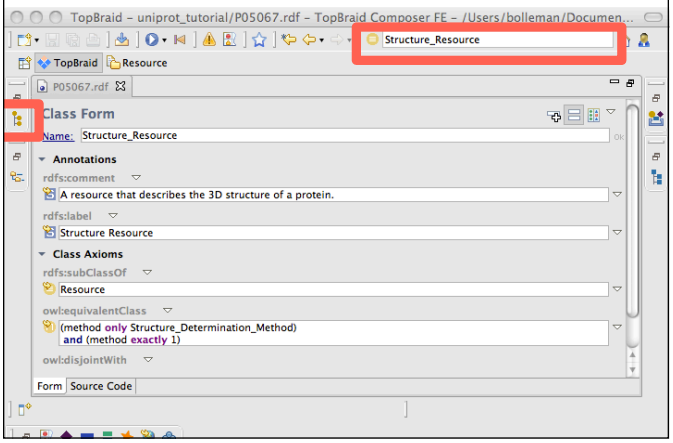
Select spin.rdf and hit ok



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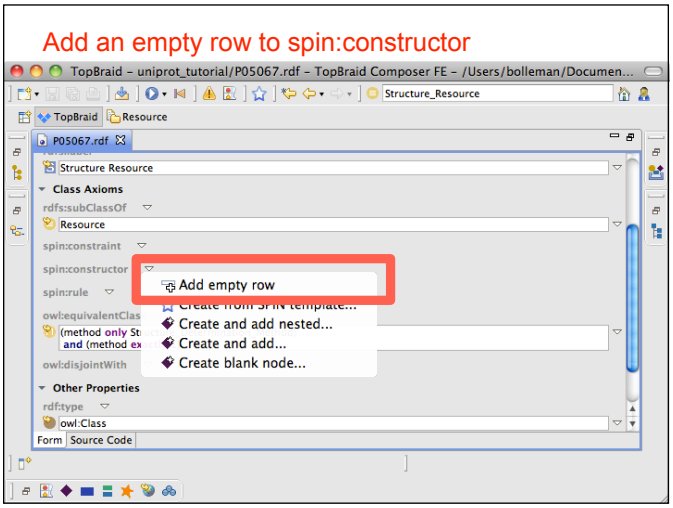
After pressing ok, save.

Structure_Resource



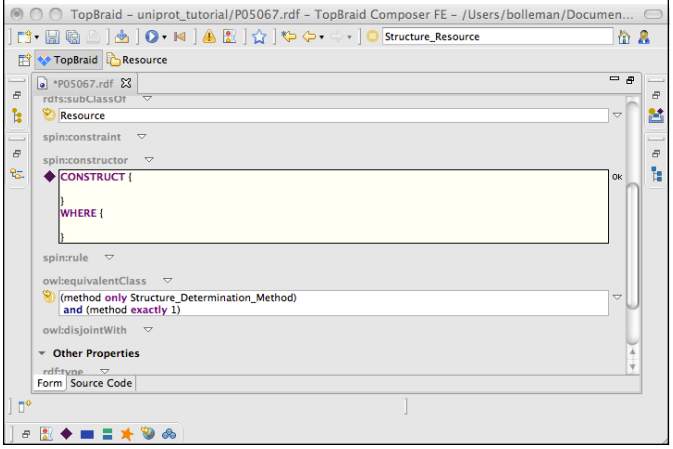
Find the Structure_Resource class. Either using the class tab or the quick navigator

Add an empty row to spin:constructor

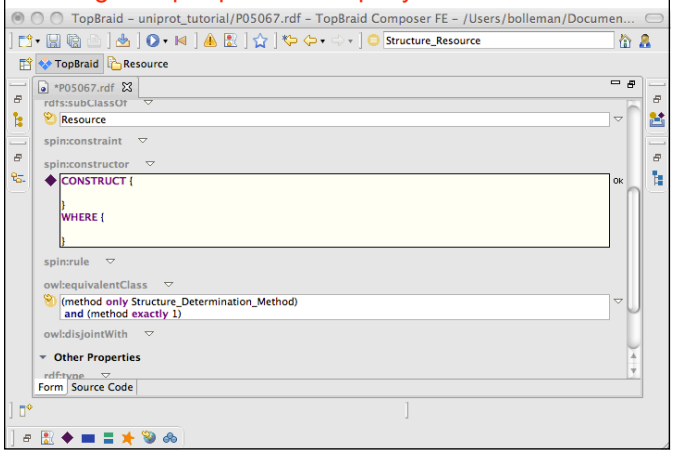


The small downwards pointing triangle next to spin:constructor is the key ui element here.

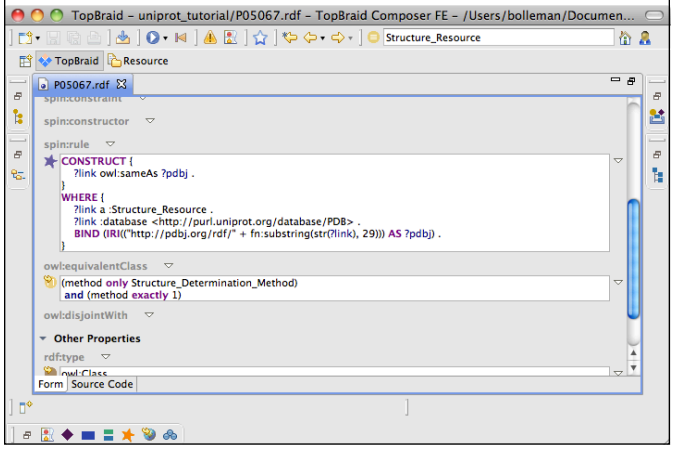
You get a sparql construct query: finish it as earlier



You get a sparql construct query finish: it as earlier



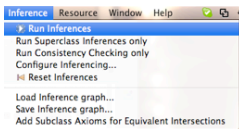
Now add the query as shown here

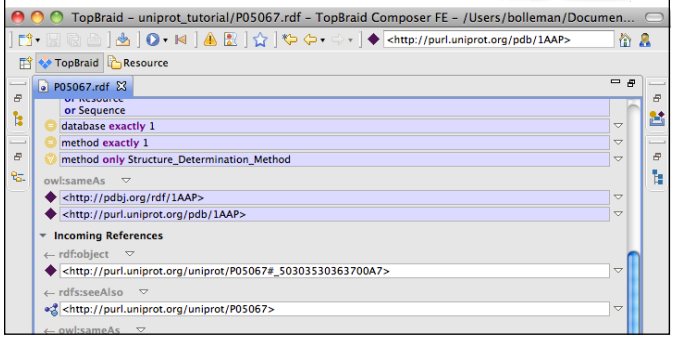


The difference is in the use of the IRI function instead of the URI function used earlier.
URI is an official synonym for the IRI function due to a small bug you can't use it here.

Run the reasoner

- In the menu "Inference" > select the option "Run inferences"





See the new owl:sameAs links. You just mapped uniprot purl identifiers with pdbj identifiers and made them logically point to the same Resource.

Running spin on lots of data without Topbraid composer

- Open Source
  - Have a look at www.spinrdf.org
- Closed Source
  - Have a look at the alegro graph triple store

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